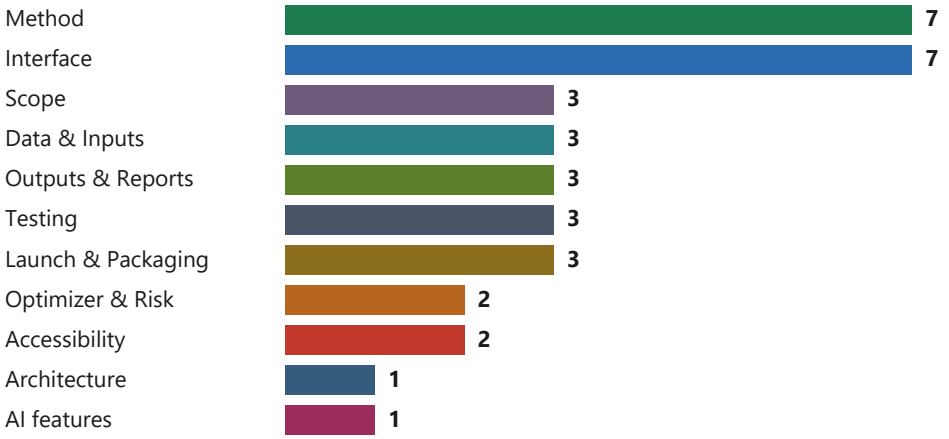
AI log: changes made to the tool

Every change made to GreenLCC India with the AI assistant, in the order it was made: what changed and why it helps. Changes to the tool only; documents and presentations are not listed.

Changes by version

Version	Date	Changes
Version 1: first build	23 Oct 2026	**15**
Version 3: simplified and aligned with CE 725	29 Oct 2026	**9**
Version 3.1: demo-ready	4-5 Nov 2026	**11**
Total		**35**

Changes by category



How to read this log

Rows are numbered in the order the changes were made and grouped by version. The coloured tag shows the **category** of each change; the **rationale** says what the change gives the user or the team.

Categories

Architecture · Scope · Method · Optimizer & Risk · Data & Inputs · AI features · Outputs & Reports · Interface · Accessibility · Testing · Launch & Packaging

Sr. no.	Date	Category	Change made	Rationale
Version 1: first build (23 Oct 2026, 15 changes)				
1	23 Oct 2026	Architecture	Built the app in Python with Streamlit and Plotly, with a custom green theme and one page per step.	Gives an interactive tool that runs in any browser and that the whole team can read and extend.
2	23 Oct 2026	Scope	Made the IGBC credit optimizer the centrepiece; solar, HVAC, glazing, lighting and water measures became options inside it.	Ties life-cycle costing to a rating decision owners actually face, and gives the demo one connected story.
3	23 Oct 2026	Method	Added one year-by-year cash flow engine, used for a single measure, for sweeps and for every simulation run.	One calculation path means every screen, report and test shows the same number.
4	23 Oct 2026	Method	Chose nominal rupee cash flows over one common 25-year study period, with re-purchases and a linear residual value.	Compares options with unequal lives fairly and keeps escalating prices realistic.
5	23 Oct 2026	Method	Entered every measure's cost as the difference from a code-minimum baseline (for example VRF price minus the split ACs it replaces).	Makes alternatives directly comparable and matches how IGBC credits are earned.
6	23 Oct 2026	Method	Fixed one timing convention: year-1 base prices, replacement prices inflated to the year bought, residual = last purchase x remaining-life fraction.	Keeps results consistent and easy to verify by hand.
7	23 Oct 2026	Optimizer & Risk	Added an optimizer that costs every legal combination (exact enumeration), applying the roof limit and the mandatory-credit rules.	Finds the provably best mix for a library of this size, and is simple to explain.
8	23 Oct 2026	Optimizer & Risk	Added break-even points, a tornado sensitivity chart, a MARR sweep and a seeded Monte Carlo simulation.	Shows how far an assumption can move before the answer changes; the fixed seed makes results repeatable.
9	23 Oct 2026	Data & Inputs	Labelled all demo prices, tariffs and IGBC point tables as sourced estimates or illustrative.	Makes the data status clear to every reader and invites users to swap in real values.
10	23 Oct 2026	AI features	Added AI reading of vendor quotes and AI polishing of the memo, each with an offline fallback (built-in parser, fixed template).	The tool works with no API key or internet, so a demo never depends on a connection.
11	23 Oct 2026	Outputs & Reports	Added a download centre: PDF decision report, Excel results, IGBC checklist, procurement shortlist, cash flow and library files.	Gives owners and consultants outputs they can use without rework.
12	23 Oct 2026	Outputs & Reports	Made the PDF decision report use the fixed template memo instead of AI-written text.	A printed report always reproduces exactly the same numbers.
13	23 Oct 2026	Interface	Reduced seven pages to Home plus four steps (Set up, Recommendation, Can we trust it?, Report); moved the target level to the sidebar; added a one-sentence answer at the top of each page, Next buttons, plain-language labels and a 'Switch target' shortcut.	The answer appears first and the flow follows the way a user thinks.

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Version 1: first build (23 Oct 2026, 15 changes)				
14	23 Oct 2026	**Interface**	Added a Quick what-if with two sliders (electricity price growth and MARR) on the Recommendation page.	Lets a user see the answer move instantly, without leaving the page.
15	23 Oct 2026	**Testing**	Added hand-calculated unit tests (annuity, replacement, residual, escalation, degradation, solar) and automatic tests of every page.	Every formula and screen is re-checked after each change.

Sr. no.	Date	Category	Change made	Rationale
Version 3: simplified and aligned with CE 725 (29 Oct 2026, 9 changes)				
16	29 Oct 2026	**Scope**	Removed inputs and outputs that were unused or duplicated: built-up area, FAR bonus, subsidy, per-measure escalation, Pareto and MACC charts, simple payback.	Leaves only inputs the maths uses, so the tool is easier to understand and to defend.
17	29 Oct 2026	**Method**	Aligned every metric with CE 725: MARR, real MARR (Fisher equation), EUAC, BCR, discounted payback, IRR/MIRR, and the headline 'return on the extra capex' (incremental IRR vs MARR).	Every number on screen maps to a course concept.
18	29 Oct 2026	**Method**	Ranked options on present worth and EUAC only; IRR and MIRR are used to accept or reject a measure or an increment.	Avoids the IRR ranking paradox (Ch 8) and keeps recommendations consistent.
19	29 Oct 2026	**Method**	Replaced the components file with a measure life plus one replaceable part, re-bought at an inflated price, with salvage credited in the final year.	Simpler inputs and a direct use of the planning-horizon method (Ch 5).
20	29 Oct 2026	**Data & Inputs**	Replaced four CSV files and a JSON file with one Excel input workbook (README, Project, Measures, IGBC_credits, IGBC_levels) with units, meanings and sources.	Users fill in one familiar file, and sources travel with the numbers.
21	29 Oct 2026	**Data & Inputs**	Rebuilt the demo from one sizing basis: a 12,000 m2 Pune office, MARR 10%, inflation 5%, tariff Rs 13/kWh, sourced equipment prices.	Realistic, consistent numbers that tell their own story without tuning.
22	29 Oct 2026	**Testing**	Added 14 CE 725 textbook examples as tests (interest factors, present worth, EUAW, IRR, multiple IRR, MIRR) and a check against closed-form geometric-gradient factors.	Shows that the finance functions match the course material.
23	29 Oct 2026	**Interface**	Added a cash flow diagram per measure and an on-screen factor-notation hand-check of one measure.	Lets the team show on screen that the engine agrees with the textbook method.

Sr. no.	Date	Category	Change made	Rationale
Version 3: simplified and aligned with CE 725 (29 Oct 2026, 9 changes)				
24	29 Oct 2026	**Launch & Packaging**	Added launcher.py (finds a free port and opens the browser) and Start_GreenLCC.bat.	One double-click starts the tool.

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Version 3.1: demo-ready (4-5 Nov 2026, 11 changes)				
25	4 Nov 2026	**Interface**	Added hover popups for 25 key terms (MARR, present worth, EUAC, Monte Carlo and more) that say what it is, why we use it and how it is used.	Explains the tool in place, so users and TAs do not need a separate glossary.
26	4 Nov 2026	**Accessibility**	Made the popups open by keyboard (Tab) and by tap, with a visible focus ring, and kept them on screen at every position.	Works for every user and on touch screens, not only with a mouse.
27	4 Nov 2026	**Scope**	Removed the AI vendor-quote import (code, sample quotes and its tests).	Keeps AI to one small, optional role (memo wording), so every number stays formula-driven and checkable.
28	4 Nov 2026	**Interface**	Home now previews the input workbook sheet by sheet with the real demo data, lists 'Things to try' and has a start-over button.	First-time users see what to provide and what to do next.
29	4 Nov 2026	**Interface**	Added a 'How it works' tab on the Recommendation page with the five steps, the formulas and the hand-check.	Puts the method in one place, ready to be explained.
30	4 Nov 2026	**Outputs & Reports**	The Report page now previews page 1 of the PDF and includes a 'Where AI is used' note.	Users see the result before downloading and know exactly what AI does in the tool.
31	4 Nov 2026	**Interface**	Reworked the first key-figure card so the headline result sits on one line.	Key figures read at a glance.
32	4 Nov 2026	**Accessibility**	Status cards now say 'above' or 'below' the MARR in words as well as colour; captions are darker; animation is reduced when the system asks for it.	Meaning never depends on colour alone, and text is easier to read.
33	5 Nov 2026	**Launch & Packaging**	The launcher now writes the live port into 'Open GreenLCC in browser.url'; added a test that compiles launcher.py and app.py.	The shortcut always opens the running app, and startup code is checked automatically.
34	5 Nov 2026	**Launch & Packaging**	Moved from a Windows .exe to a localhost-only package: pinned requirements, the optional AI package in requirements-ai.txt, start scripts for Windows and Mac/Linux and a START_HERE.txt.	Reviewers run the tool on their own PC with a predictable, repeatable install.

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Version 3.1: demo-ready (4-5 Nov 2026, 11 changes)				
35	5 Nov 2026	Testing	Checked every page in a real Chrome browser (about 2 seconds per page, no errors) and re-ran all 82 tests from a fresh install of the zip.	Confirms the delivered package works exactly as shipped.